Hadoop & Spark - ISEN 2025/2026

Lab: Spark Advanced Functions

For this Lab you will need:

- The 3.10 Lab

As a reminder

Check 3.1 to login in NGROK, Kaggle, install Java or Spark and import libraries.

UDFs

For this part, you will need the Cabs.csv in the data section of this course.

Just drag and drop with this:

```
from google.colab import files
files.upload()
!ls
```

UDFs

Let's read the dataset:

```
# Read Cabs file
cabsDF = (
              spark
                .read
                .option("header", "true")
                .option("inferSchema", "true")
                .csv("Cabs.csv")
         )

# Create temp view
cabsDF.createOrReplaceTempView("Cabs")

cabsDF.show(truncate=False)
```

UDFs

Let's take a look at the `Name` column:

```
cabsDF.select("Name").show()
```

Everything is formatted correctly, but everything is in uppercase. Let's try to change that with a UDF following those rules:

- We want An uppercase at the beginning of every word, the other letters in lowercase
- Delete the coma

UDFs

First we declare our function that will be applied to each row.

```
import re

def convertCase(string):
    result = ""

    # Split the input string into words, considering both spaces and commas as separators
    name_words_array = re.split('[ ,]', string)

    for name_word in name_words_array:
        if name_word:  # Check if the name_word is not empty
            result += name_word[0].upper() + name_word[1:].lower() + " "

    # Remove the trailing space and return the result
    return result.strip()
```

UDFs

Declare the udf:

```
convertCaseUdf = udf( lambda str: convertCase(str), StringType())
```

Apply it and see the result:

```
(
    cabsDF
        .select(
                    "Name",
                    convertCaseUdf(col("Name")).alias("Name_ConvertedCase")
                )
).show(truncate=False)
```

UDFs

You can also do it like this:

```
(
    cabsDF
      .withColumn ("Name_ConvertedCase", convertCaseUdf(col("Name")))
      .select("Name","Name_ConvertedCase")

).show(truncate=False)
```

UDFs

You can do the exact same thing in SQL:

```
spark.udf.register("convertCaseSqlUdf", convertCase, StringType())
```

And apply it:

```
spark.sql("""
    SELECT Name
         , convertCaseSqlUdf(Name) AS Name_ConvertedCase
    FROM Cabs
""").show(truncate=False)
```

Exercise UDF

Let's try to change that following those rules:

- We want An uppercase at the beginning of every FIRST word, the other letters in lowercase
- If there is a coma, the next letter will be uppercase

Ex:

- `GAUTHIER,JACQUES` -> `Gauthier, Jacques`
- `BOULOS TAXI CORP.` -> `Boulos taxi corp.`

> Hint: make sure you replace the instances of ", "
> (one coma followed by a space) by ","

Windows Functions

Preparation

As usual, let's load the Taxi dataset if not already done:

```
!kaggle datasets download -d microize/newyork-yellow-taxi-trip-data-2020-2019
!unzip newyork-yellow-taxi-trip-data-2020-2019.zip
```

Preparation

```
yellowTaxiSchema = (
                        StructType
                        ([
                            StructField("VendorId"               , IntegerType()   , True),
                            StructField("PickupTime"             , TimestampType() , True),
                            StructField("DropTime"               , TimestampType() , True),
                            StructField("PassengerCount"         , DoubleType()    , True),
                            StructField("TripDistance"           , DoubleType()    , True),
                            StructField("RateCodeId"             , DoubleType()    , True),
                            StructField("StoreAndFwdFlag"        , StringType()    , True),
                            StructField("PickupLocationId"       , IntegerType()   , True),
                            StructField("DropLocationId"         , IntegerType()   , True),
                            StructField("PaymentType"            , IntegerType()   , True),
                            StructField("FareAmount"             , DoubleType()    , True),
                            StructField("Extra"                  , DoubleType()    , True),
                            StructField("MtaTax"                 , DoubleType()    , True),
                            StructField("TipAmount"              , DoubleType()    , True),
                            StructField("TollsAmount"            , DoubleType()    , True),
                            StructField("ImprovementSurcharge"   , DoubleType()    , True),
                            StructField("TotalAmount"            , DoubleType()    , True),
                            StructField("CongestionSurcharge"    , DoubleType()    , True),
                        ])
                    )
```

Preparation

Read and create temp view:

```
yellowTaxiDF = (
                  spark
                    .read
                    .option("header", "true")
                    .schema(yellowTaxiSchema)
                    .csv("yellow_tripdata_2020-04.csv")
                )
yellowTaxiDF.createOrReplaceTempView("YellowTaxis")
yellowTaxiDF.printSchema()
```

Preparation

Then load the taxi zones:

```
taxiZonesSchema = "PickupLocationId INT, Borough STRING, Zone STRING, ServiceZone STRING"
taxiZonesDF = (
                  spark
                    .read
                    .schema(taxiZonesSchema)
                    .csv("taxi+_zone_lookup.csv")
               )
taxiZonesDF.createOrReplaceTempView("TaxiZones")
taxiZonesDF.printSchema()
```

Windows - Find share of each borough in terms of rides

Let's get the rides count in each borough:

```
taxiRidesDF = (

  spark.sql("""
                SELECT tz.Borough
                     , COUNT(*)     AS RideCount

                FROM TaxiZones tz
                    INNER JOIN YellowTaxis yt ON yt.PickupLocationId = tz.PickupLocationId

                GROUP BY tz.Borough
   """)
)

taxiRidesDF.createOrReplaceTempView("TaxiRides")

taxiRidesDF.orderBy("Borough").show()
```

Windows - Find share of each borough in terms of rides

Calculate total rides across all boroughs by:

- 1 Creating a Window over entire table
- 2 Adding total rides (across all boroughs) against each row

```
taxiRidesWindowDF = (
    spark.sql("""
                    SELECT *
                         , SUM (RideCount)   OVER ()   AS TotalRideCount

                    FROM TaxiRides
    """)
)

taxiRidesWindowDF.orderBy("Borough").show()

taxiRidesWindowDF.createOrReplaceTempView("TaxiRidesWindow")
```

Windows - Find share of each borough in terms of rides

Let's find the share of each borough in terms of rides by dividing Borough's ride count with Total Ride Count (across all boroughs)

```
(spark.sql("""

            SELECT *
                  , ROUND( (RideCount * 100) / TotalRideCount, 2)   AS RidesSharePercent

            FROM TaxiRidesWindow
            ORDER BY Borough

""")).show()
```

Windows - Find share of each borough in terms of rides

Get rides for each zone , with Zone being part of Borough

```
taxiRidesDF = (
    spark.sql("""
                    SELECT tz.Borough
                         , tz.Zone
                         , COUNT(*)     AS RideCount

                    FROM TaxiZones tz
                        INNER JOIN YellowTaxis yt ON yt.PickupLocationId = tz.PickupLocationId

                    GROUP BY tz.Borough
                           , tz.Zone
    """)
)

taxiRidesDF.orderBy("Borough", "Zone").show(truncate=False)

taxiRidesDF.createOrReplaceTempView("TaxiRides")
```

Windows - Find share of each borough in terms of rides

Let's calculate the total rides across each borough by

- 1 Creating Window over entire table, and partitioning by Borough
- 2 Adding total rides (across all zones in a borough) against each row

```
taxiRidesWindowDF = (
  spark.sql("""

            SELECT *
                 , SUM (RideCount) OVER (PARTITION BY Borough) AS TotalRideCountByBorough
            FROM TaxiRides
  """)
)

taxiRidesWindowDF.orderBy("Borough", "Zone").show(truncate=False)

taxiRidesWindowDF.createOrReplaceTempView("TaxiRidesWindow")
```

Windows - Find share of each borough in terms of rides

Let's find shares of each zone in terms of rides, within their borough by dividing Zone's ride count with Borough Ride Count:

```
(spark.sql("""

    SELECT *
          , ROUND( (RideCount * 100) / TotalRideCountByBorough, 2)   AS RidesSharePercentInBorough

    FROM TaxiRidesWindow
    ORDER BY Borough, Zone

""")).show(truncate=False)
```

Exercice netflix Shows

Let's use the "Netflix Movies and TV Shows" dataset available on Kaggle. The dataset contains various details about movies and TV shows available on Netflix. You can see the dataset from here.

The objective of this exercise is to analyze the number of movies and TV shows produced by each country and calculate the percentage share of each country in terms of content production.

Let's import it and take a look.

```
!kaggle datasets download -d shivamb/netflix-shows
!unzip netflix-shows.zip
```

Exercice netflix Shows

```
netflix_schema = StructType([
    StructField("show_id", StringType(), True),
    StructField("type", StringType(), True),
    StructField("title", StringType(), True),
    StructField("director", StringType(), True),
    StructField("cast", StringType(), True),
    StructField("country", StringType(), True),
    StructField("date_added", StringType(), True),
    StructField("release_year", StringType(), True),
    StructField("rating", StringType(), True),
    StructField("duration", StringType(), True),
    StructField("listed_in", StringType(), True),
    StructField("description", StringType(), True)
])

netflix_df = spark.read.csv("netflix_titles.csv", header=True, schema=netflix_schema)
netflix_df.createOrReplaceTempView("Netflix")
netflix_df.printSchema()
```

Exercice netflix Shows

When showing the dataframe, as you can see, the `country` column contains some lists with comas (ex: `Peru, United States, China`).

show_id	type	title	director	cas	country	date_added	release_year	rating	duration	listed_in	description
s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	nul	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nea...
s2	TV Show	Blood & Water	null	Ama Qamata, Khosi..	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV ...	After crossing pa...
s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tr..	null	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, I...	To protect his fa...
s4	TV Show	Jailbirds New Orl...	null	nul	null	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reali...	Feuds, flirtation...
s5	TV Show	Kota Factory	null	Mayur More, Jiten..	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV ...	In a city of coac...
s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach..	null	September 24, 2021	2021	TV-MA	1 Season	TV Dramas, TV Hor...	The arrival of a ...
s7	Movie	My Little Pony: A...	Robert Cullen, Jo...	Vanessa Hudgens, ..	null	September 24, 2021	2021	PG	91 min	Children & Family...	Equestria's divid...
s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oya..	United States, Gh...	September 24, 2021	1993	TV-MA	125 min	Dramas, Independe...	On a photo shoot ...
s9	TV Show	The Great British...	Andy Devonshire	Mel Giedroyc, Sue..	United Kingdom	September 24, 2021	2021	TV-14	9 Seasons	British TV Shows,...	A talented batch ...
s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy,..	United States	September 24, 2021	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting...
s11	TV Show	Vendetta: Truth, ...	null	nul	null	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, D...	"Sicily boasts a ...
s12	TV Show	Bangkok Breaking	Kongkiat Komesiri	Sukollawat Kanaro..	null	September 23, 2021	2021	TV-MA	1 Season	Crime TV Shows, I...	Struggling to ear...
s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jann..	Germany, Czech Re...	September 23, 2021	2021	TV-MA	127 min	Dramas, Internati...	After most of her...
s14	Movie	Confessions of an...	Bruno Garotti	Klara Castanho, L..	null	September 22, 2021	2021	TV-PG	91 min	Children & Family...	When the clever b...
s15	TV Show	Crime Stories: In...	null	nul	null	September 22, 2021	2021	TV-MA	1 Season	British TV Shows,...	Cameras following...
s16	TV Show	Dear White People	null	Logan Browning, B..	United States	September 22, 2021	2021	TV-MA	4 Seasons	TV Comedies, TV D...	"Students of colo...
s17	Movie	Europe's Most Dan...	Pedro de Echave G...	nul	null	September 22, 2021	2020	TV-MA	67 min	Documentaries, In...	Declassified docu...
s18	TV Show	Falsa identidad	null	Luis Ernesto Fran..	Mexico	September 22, 2021	2020	TV-MA	2 Seasons	Crime TV Shows, S...	Strangers Diego a...
s19	Movie	Intrusion	Adam Salky	Freida Pinto, Log..	null	September 22, 2021	2021	TV-14	94 min	Thrillers	After a deadly ho...
s20	TV Show	Jaguar	null	Blanca Suárez, Iv..	null	September 22, 2021	2021	TV-MA	1 Season	International TV ...	In the 1960s, a H...

only showing top 20 rows

Exercice netflix Shows

As a first step we will create a new dataframe.

We can use the split() function to split the countries by commas and then explode() function to create new rows for each country. Here's how you can do it:

- 1 First, let's modify the initial netflix_df by adding a new column with the countries split by commas

> Hint: Use the split() functions seen in a previous lab

- 2 Then, use the explode() function to create new rows for each country in the split_country column

  Ex: `exploded_df = netflix_df.select("show_id", "type", "title", "director", "cast", explode("split_country").alias("country"), "date_added", "release_year", "rating", "duration", "listed_in", "description")`

- 3 with this new dataframe, get the content count by country, in a column called content_count
- 4 Create a Window over the entire table to calculate the total content count
- 5 Calculate the share of each country in terms of content production